



FACULTY:

FACULTY OF ICT

PROGRAMME:

PROGRAM

YEAR:

2

LECTURER:

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FROM:

2420004

COURSE:

OOP

COURSE CODE:

BSCICT 2203

Code

```
NS.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Student
5  {
6  private:
7      int age;
8
9  public:
10     // Default constructor
11     Student()
12     {
13         age = 18;
14         cout << "Default constructor called" << endl;
15     }
16
17     void display()
18     {
19         cout << "Age: " << age << endl;
20     }
21 };
22
23 int main()
24 {
25     Student s1; // Automatically calls default constructor
26     s1.display();
27
28     return 0;
29 }
```

The program demonstrates the implementation of a default constructor in C++. A class named `Student` is created with a private variable called `age`. Inside the public section, a default constructor `Student()` is defined, which automatically assigns the value 18 to `age` whenever an object of the class is created. The constructor also displays a message showing that it has been called. A `display()` function is included to print the value of `age`. In the `main()` function, an object `s1` is created, which automatically triggers the default constructor, and then the `display()` function is called to show the stored age value.

OUTPUT

```
C:\Users\Kamwai\Desktop\ED\Year 2\OOP\As\NS.exe
Default constructor called
Age: 18

-----
Process exited after 0.382 seconds with return value 0
Press any key to continue . . .
```

The output first displays the message "Default constructor called" because the constructor executes automatically when the object s1 is created. After that, the program prints "Age: 18" from the display() function, confirming that the constructor successfully initialized the age variable with the default value.